

SquaredLabs

Power Perpetuals: Pioneering the Future of Leveraged Trading in DeFi

Whitepaper v1.0

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1 - Abstract

Derivatives play a crucial role in traditional financial markets and are emerging as a vibrant sector in Decentralized Finance (DeFi). Recently, several projects have introduced decentralized derivative products such as options, futures, and perpetuals. Among these, continuously funded power perpetuals represent a unique innovation. Originally conceptualized as "Power Claims" by Wayne Nilsen, power perpetuals have quickly captured the interest of the crypto research community. Recognizing this potential, Squared Labs aims to pioneer an ecosystem of products and services built upon the foundation of perpetual power derivatives. By offering leveraged long and short exposure without liquidation risks, we empower traders to navigate the market with entirely new strategies and capitalize on lucrative opportunities. Power perpetuals have the potential to consolidate significant liquidity from the options market into a single, capital-efficient instrument.

2 - Introduction to Power Derivatives

2.1 - Defining Power Perpetuals: Leveraged Exposure Without Limits

A power perpetual is a perpetual derivative indexed to a power p of the price of an underlying instrument. Given a price x of an asset such as *BTC*, the value of the *BTC* power perpetual is:

$$BTC^p = x^p, p \geq 1$$

The mark price of the power perpetual is kept in line with the index price through a funding fee mechanism:

$$Funding\ fee = Mark - Index = Mark - x^p$$

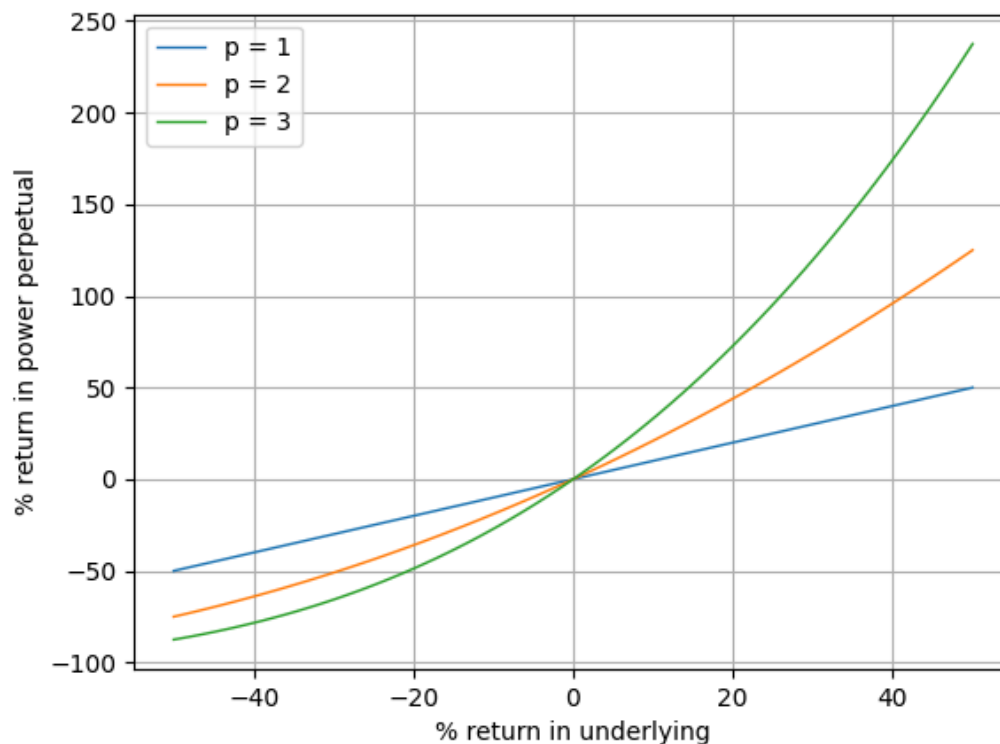
The funding mechanism plays a crucial role in keeping the power perpetual's mark price aligned with the underlying asset's index price. When the mark price surpasses the index price, long positions are required to pay a funding fee to short positions. On the other hand, if the index price exceeds the mark price, short positions must pay the funding fee to long positions. This system creates an incentive for traders to take positions that push the mark price closer to the index price, effectively reducing deviations. As a result, the funding mechanism ensures that the power perpetual maintains its intended exposure to the underlying asset, preventing significant price disparities and promoting efficient market dynamics.

By providing leveraged exposure without the constraints of strikes or expiration dates, power perpetuals have the potential to consolidate a significant portion of options market liquidity into a single, capital-efficient instrument. This innovative derivative product offers traders a streamlined way to gain amplified exposure to price movements in the underlying asset, while reducing the fragmentation of liquidity across multiple strike prices and expiration dates typically seen in traditional options markets.

2.2 - Key Characteristics and Benefits of Power Perpetuals

Continuously funded power perpetuals have no expiration and provide positive convexity. This means favorable price movements in the underlying asset yield greater positive returns compared to the absolute negative returns from equally sized unfavorable moves. In other words, power perpetual holders earn more when the price moves in their favor than they lose when it moves against them.

For cases where $p \geq 1$, the power perpetual return profile exhibits heightened sensitivity to the underlying return.

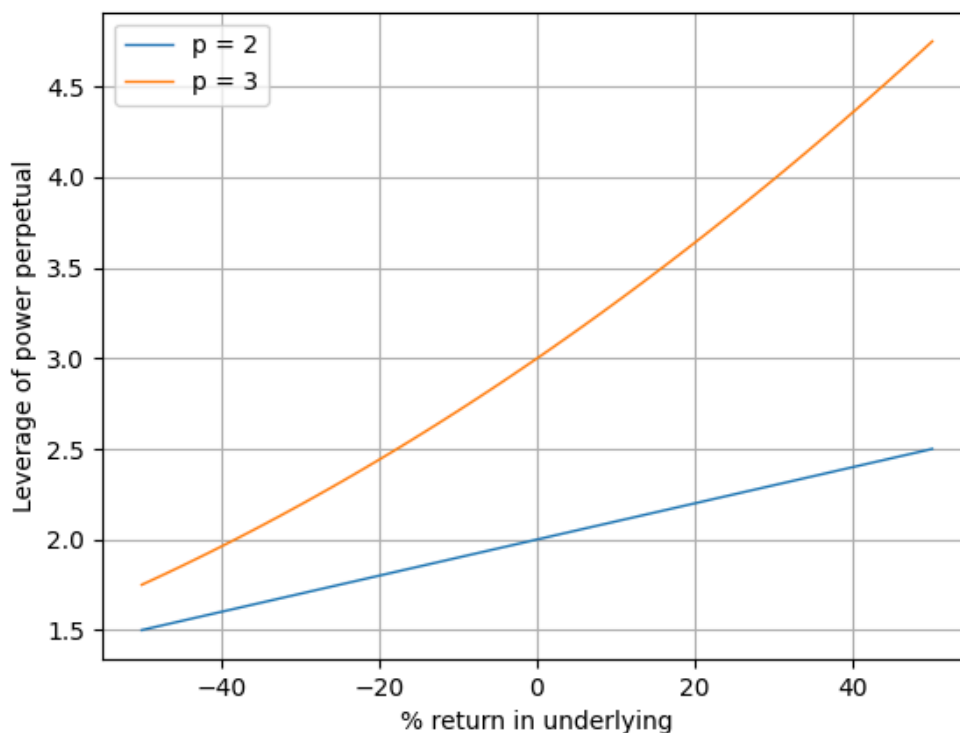


As the return of the underlying asset rises above zero, the power perpetual's return increases exponentially, effectively providing traders with built-in leverage. This unique characteristic distinguishes power perpetuals from traditional leveraged products, as it allows traders to potentially amplify their returns without exposing them to liquidation risk. By design, the power

perpetual's value (x^p) always remains positive, ensuring that traders' positions are shielded from liquidation, even during periods of extreme market volatility.

The absence of liquidation risk is a key advantage of power perpetuals, making them particularly attractive to traders seeking to manage risk while still having the potential for enhanced returns. This feature allows traders to maintain their positions through market turbulence, providing greater flexibility and control over their trading strategies. As a result, power perpetuals offer a compelling alternative to traditional leveraged products, combining the potential for amplified returns with a more robust risk management framework.

This setup enables more efficient capital allocation when trading with full margin. The minimum leverage, approximately equal to power p , is experienced for small returns around 0. Leverage increases exponentially with positive returns while diminishing with negative returns.



Shorting perpetual futures is traditionally associated with collateral liquidation risk. To address this

fundamental problem and eliminate aforementioned risk Squared Labs introduces a novel bonding curve mechanism to be discussed in the implementation paper.

2.3 - Illustrative Scenarios: Quantifying the Power Perpetual Advantage

The return of a power perpetual r_{pp} can be calculated as

$$r_{pp} = (1 + r_u)^p - 1$$

where r_u is the return of the underlying asset and p is the power.

Scenario 1

BTC price moves from 60.000\$ to 66.000\$, a 10% return $r_u = 0.1$. The return of BTC^2 is:

$$r_{pp} = (1 + 0.1)^2 - 1 = 0.21 = 21\%$$

Scenario 2

BTC price moves from 60.000\$ to 54.000\$, a -10% return $r_u = -0.1$. The return of BTC^2 is:

$$r_{pp} = (1 + (-0.1))^2 - 1 = -0.19 = -19\%$$

These examples illustrate how traders earn more when the price moves favorably than they lose when it moves unfavorably. For the same 10% underlying move, a BTC^2 trader earns 21% on the upside but only loses 19% on the downside.

Scenario 3

USDT price moves from 0.995\$ to 1.005\$, a +1.005% return $r_u = 0.01005$. The return of $USDT^3$ is:

$$r_{pp} = (1 + 0.01005)^3 - 1 = 0.0345 = 3.0454\%$$

Assuming *USDT* trades at 0.995, $USDT^3$ would trade at 0.9850. A trader investing \$10,000 into a $USDT^3$ long position at \$0.9850 would receive 10,151.51 $USDT^3$ tokens. If *USDT* later trades at

\$1.005 and $USDT^3$ at \$1.015, the trader could cash out for \$10,304.54, netting a \$304.54 (3.045%) profit.

3 - Squared Labs: Empowering Traders Through Permissionless Power Derivatives

3.1 - Democratizing Access: Creating Power Perpetuals for Any Asset

Squared Labs aims to democratize the power perpetual market by allowing users to create permissionless power perpetual products for their desired underlying assets.

Launching such products has traditionally been limited to assets with significant liquidity in secondary derivatives markets. However, by combining power bonding curves with existing pricing technologies, Squared Labs has largely automated the processes related to issuance and risk management.

3.2 - Customizable Convexity: Tailoring Risk and Return with Adjustable Power

To cater to diverse market participants, a key feature will be allowing users to customize the power p of their perpetual derivative. By selecting p , users can effectively tune their leverage and convexity according to their needs.

This customization enables a wide range of power derivative products suitable for complex trading strategies, short-term speculative hedges, or simply vehicles for profit.

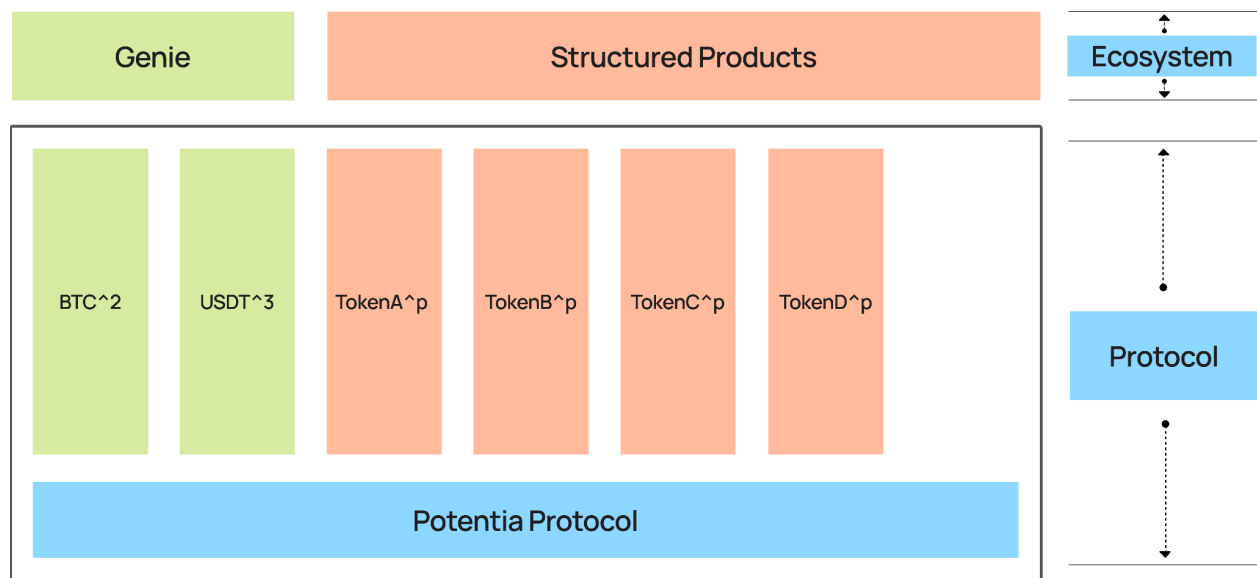
3.3 - Diverse Product Suite: From Hedging to Speculation

Squared Labs offers a diverse suite of power perpetual products catering to various trading objectives and strategies. Whether users seek to hedge against short-term market volatility, speculate on asset prices, or employ sophisticated algorithmic trading techniques, our platform provides the necessary tools and flexibility.

By offering a range of power perpetual instruments with varying convexity levels, we enable users to fine-tune their risk exposure and optimize their returns based on their market views and risk appetites. This empowers traders to execute their strategies with greater precision and efficiency.

4 - The Squared Labs Ecosystem

Squared Labs ecosystem is a set of services built on top of Potentia protocol. Potentia allows crypto token issuers to create derivative markets for tokens they issue. Our portfolio of products and services built on top of Potentia goes under brand name Genie. It provides fundamental market facilities such as DEX trading and liquidity provisioning. Additionally it is a platform for showcasing applications and utility of power derivatives and Potentia. As an open protocol Potentia allows any centralized or decentralized applications to leverage built in market making and risk management features . Creating complex trading products, conservative hedging strategies or high risk leveraged products is at disposal for any party integrating with the protocol.



Block diagram above gives an overview of Potentia protocol and its relationship with the ecosystem. The protocol can spawn power perpetual pools of various powers and underlying assets. On top of the protocol any 3rd party can build user facing services. In a permissionless environment anyone can extend the protocol by building out structured products as a part of the ecosystem.

4.1 - Power Pools: Permissionless Liquidity Provisioning

Power Pool (PP) is the core protocol component that enables liquidity providers to create permissionless power derivative liquidity pools and manage them. For end users PPs enable seamless DEX style trading of power derivatives against underlying assets.

Simple Pools: Single-Asset Power Perpetuals

Simple Pools contain a single power perpetual of a specified power (e.g., BTC2_BTC). These pools provide a straightforward way for liquidity providers to offer power perpetual trading for individual assets.

Composite Pools: Multi-Asset Power Perpetual Families

Composite Pools contain a family of power perpetual products (e.g., BTC2_BTC, BTC3_BTC, BTC4_BTC) sharing liquidity in the underlying asset. This allows for more efficient capital utilization and provides traders with a variety of convexity options within a single pool.

To create a power perpetual (PP), liquidity providers are required to pay an activation fee in SQL tokens and lock up a protocol-specified minimum requirement quantity (MRQ) of SQL. The activation fee and MRQ are designed to ensure that liquidity providers have sufficient skin in the game and are committed to the long-term success of the pool. The total amount of SQL locked by the liquidity provider determines the profit-sharing ratio between the Squared Labs protocol and the pool owner.

If the MRQ increases beyond the owner's locked quantity, their profits are automatically redirected to purchase additional SQL until the locked amount reaches the new MRQ. This mechanism incentivizes pool owners to maintain sufficient SQL holdings and align their interests with the protocol. Pool owners have the option to terminate their PPs, but it's important to note that their SQL remains locked until the termination period expires, further encouraging long-term commitment to the pool's success.

4.2 - Structured Products: Advanced Trading Tools and Strategies

Structured products offer API and UI access to Squared Labs' offerings for traders and DeFi builders, including:

Short-Term Hedges: Navigating Market Volatility

Power perpetuals enable efficient short-term hedging strategies to protect against anticipated market volatility. By leveraging the convexity of these instruments, traders can safeguard their portfolios from temporary downturns or capitalize on expected short-term price movements without risk of liquidation.

Automated Trading: Harnessing the Power of Algorithms

Squared Labs provides tools and infrastructure for automated trading strategies involving power perpetuals. Traders can design and deploy custom algorithms to capitalize on market inefficiencies, execute complex trading patterns, or maintain efficient risk management practices.

Perpetual Routing: Seamless Cross-Asset Trading

Perpetual routing allows power pool owners to facilitate the trading of their power perpetuals against any other asset or derivative within the Squared Labs protocol. This enables traders to seamlessly move between different power perpetuals and other assets, unlocking new arbitrage opportunities and enhancing overall market liquidity.

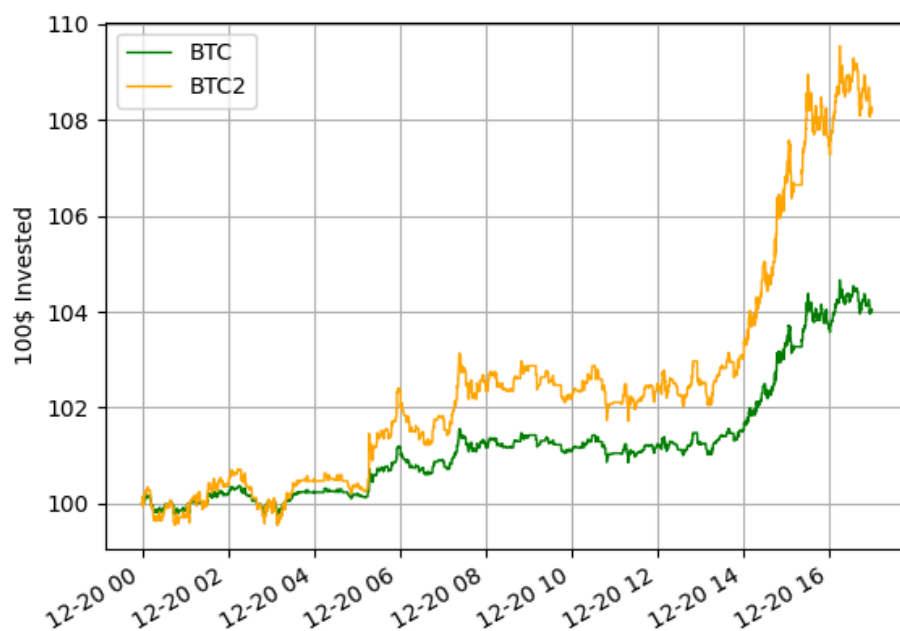
4.3 - Flagship Offerings: BTC² and USDT³

Bitcoin (BTC) and Tether (USDT) have been crucial to the crypto ecosystem since their inception. Understanding the practical applications of their power derivatives in real-world trading is the focus of Squared Labs' initial product offering.

BTC²: Amplifying Bitcoin's Volatility

BTC² offers quadratic price exposure to Wrapped Bitcoin (WBTC), providing traders with the potential for amplified returns without the use of traditional leverage. As Bitcoin's price changes, BTC² tracks the square of the WBTC price, resulting in a return profile that mimics the effects of trading with 2x leverage. However, unlike leveraged positions, BTC² holders' positions are protected from liquidation. This unique feature allows traders to capitalize on the amplified movements of Bitcoin's price while mitigating the risks typically associated with leveraged trading.

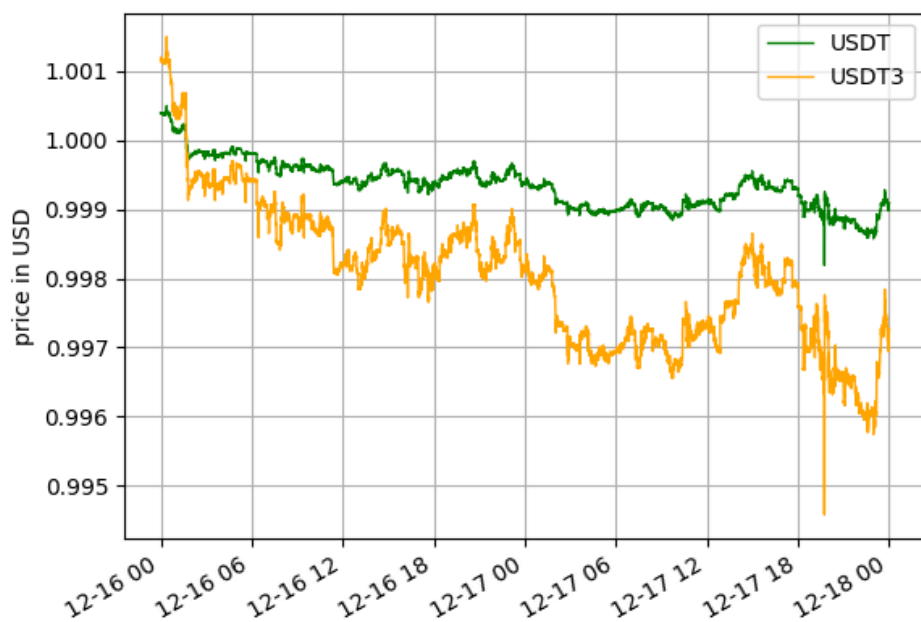
Only a Bitcoin price of exactly zero could cause a total loss—an extremely improbable scenario. Traders anticipating downward Bitcoin moves can lock collateral and short BTC².



USDT³: Capturing Stablecoin Fluctuations

USDT³ represents a novel extension of Tether, significantly amplifying price changes with the cube of the underlying price. This allows capturing even the slightest changes in USDT value, opening new avenues for short-term trading and hedging strategies.

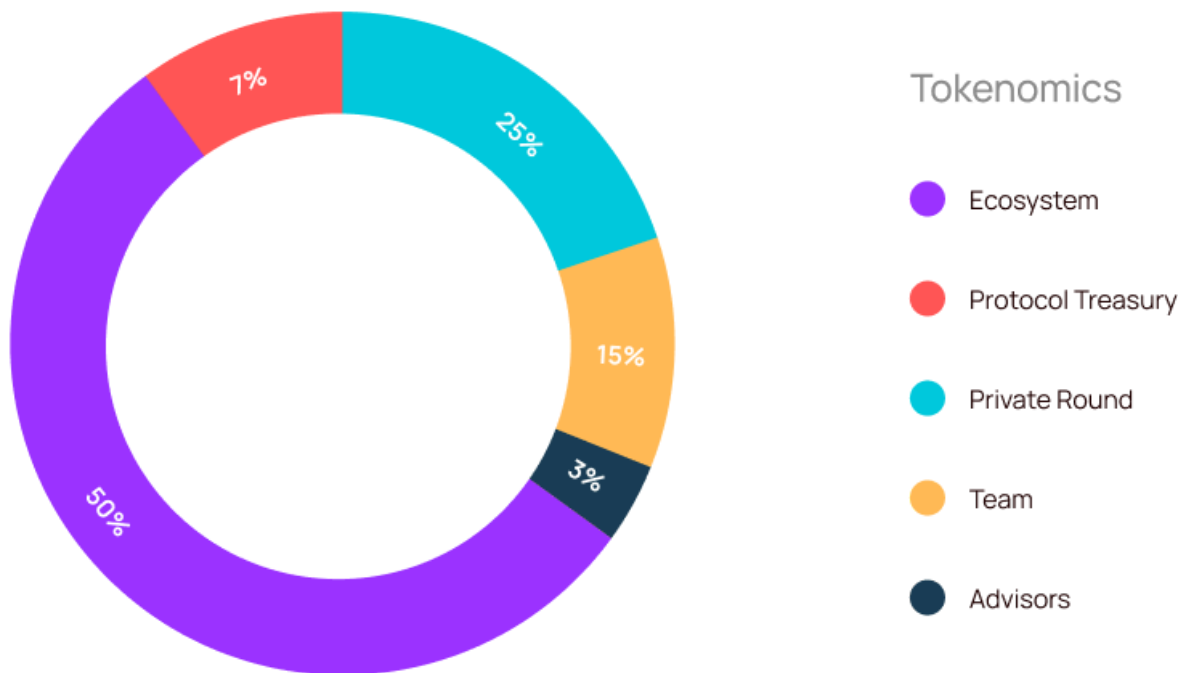
The graph below compares the standard USDT price (green) with its cubic transformation, USDT^3 , over a 3-day period, illustrating the latter's pronounced response to market volatility.



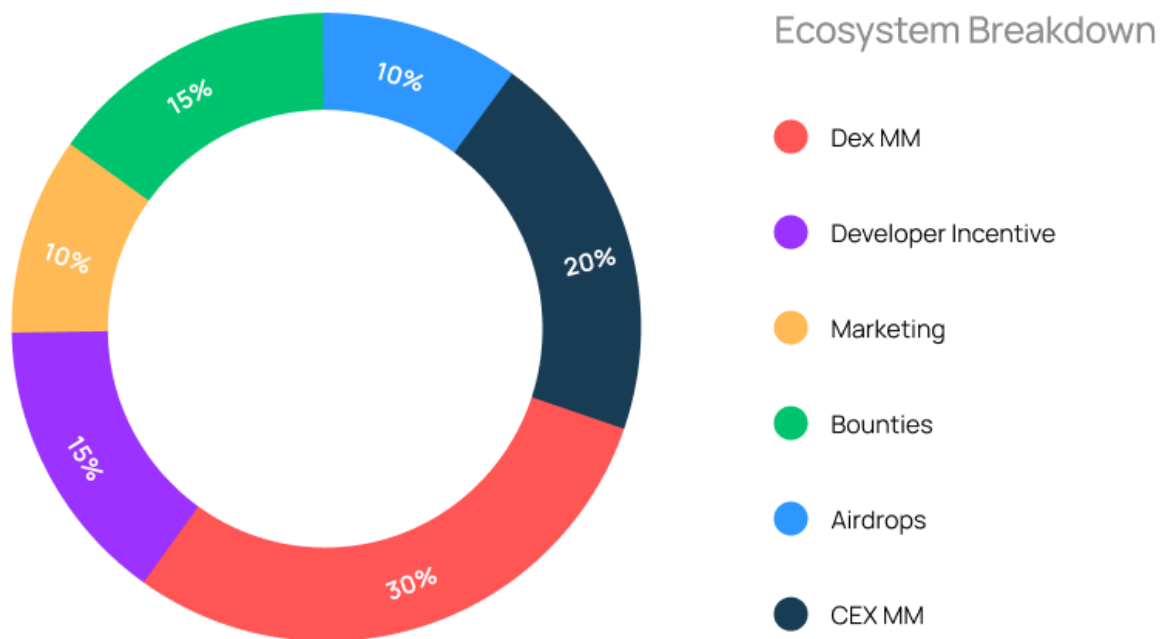
5 - SQL Token: Fueling the Squared Labs Economy

5.1 - Token Utility: Empowering Users and Driving Adoption

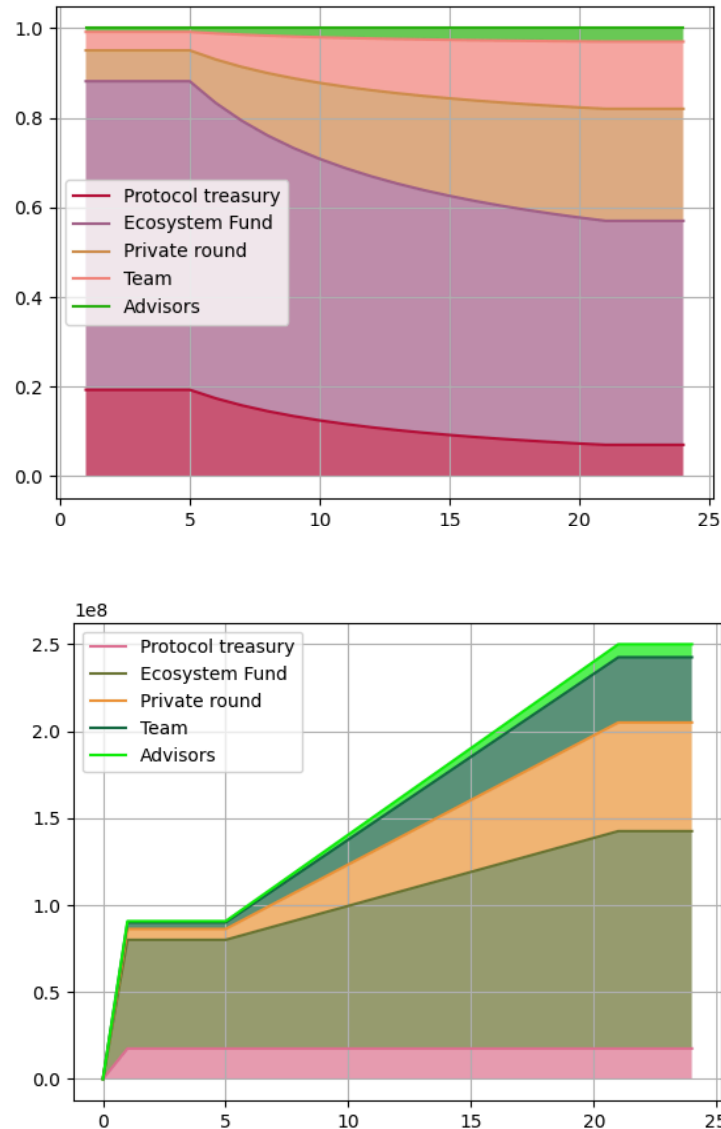
The SQL token is Squared Labs' platform and protocol token with a maximum supply of 250M, allocated as follows:



The ecosystem development fund incentivizes builders, liquidity providers, and traders to participate in the protocol and drive innovation. Prior to the token generation event (TGE), 50% of ecosystem funds will be offered through a Variable Rate Gradual Dutch Auction (VRGDA). The remaining funds will be allocated according to this schedule:



Pre-TGE tokens subject to vesting will be 10% unlocked at TGE, with the remainder vested over 21 months following a 6-month cliff.



As a protocol, Squared Labs serves various participants by offering products and services built upon power derivatives. Basic retail-oriented services like trading and automated position management will not require users to hold or lock SL tokens.

However, accessing advanced protocol features may require paying fees in SQL or locking tokens during usage, depending on the product or service. For example, automated trading and hedging strategies through the SquaredLabs Genie interface will incur usage fees. Advanced users operating their own power perpetual markets must pay a one-time fee to create a market and keep a specified SQL quantity locked while the power pool is active.

5.2 - Governance: Balancing Protocol Control and Community Participation

Squared Labs, the owner and facilitator of the protocol and services, maintains full governance control. This includes activating and deactivating protocol contracts and services and setting base protocol fees.

Pool operators can permissionlessly create power pools and provide services to their users. Squared Labs cannot prevent third parties from using its permissionless services. However, accessing the full range of services, such as custody and insurance, requires KYC, and Squared Labs reserves the right to refuse service.

Squared Labs recognizes the community's need to shape the protocol. To facilitate user participation in decision-making, Squared Labs will actively create user pools and referendums. Participants in the voting process will be rewarded with airdrops from the ecosystem fund proportional to their voting power:

$$V = Q^{1/p}$$

where:

V is voting power

Q is the user's SQL token quantity

p is the power factor

The power factor penalizes whales. Squared Labs will maintain this factor at or above 2 to prevent whales from dominating the voting process.

6 - Future Roadmap: Expanding the Power Perpetual Frontier

Squared Labs is committed to expanding its offerings and driving the adoption of power perpetuals. Key milestones on our roadmap include:

Q3 2024

- Launch additional power perpetuals for top 10 cryptocurrencies
- Potentia Mainnet
- Introduce composite pools for enhanced capital efficiency
- Integrate with major DEXs and CEXs for increased liquidity

Q4 2024

- Release Genie v1.0 for automated trading and hedging
- Offer DAO and structured product treasuries access to power perpetuals
- Partner with leading DeFi insurance protocols

Q1 2025

- Expand to power perpetuals for major DeFi tokens
- Launch cross-chain deployments on leading Layer 1 and Layer 2 networks
- Introduce governance staking and user-generated power pools

Q2 2025 and Beyond

- Enable power perpetuals for selected real-world assets (commodities, equities, FX)
- Decentralize protocol governance through progressive transfer to SQL token holders
- Continuously innovate and launch new structured products

7 - Conclusion: Shaping the Evolution of DeFi Derivatives

Permissionless power perpetuals have the potential to revolutionize the trading landscape by providing unique tools for traders to capitalize on or hedge against price movements in various underlying assets. These innovative products are likely to attract sophisticated traders and institutions, leading to increased trading volumes and enhanced liquidity. As more participants enter the power perpetual market, network effects may emerge, further amplifying the utility and value of the Squared Labs ecosystem.

By adhering to the principles of DeFi composability and facilitating novel trading and hedging strategies, Squared Labs' products will contribute to greater market efficiency and price discovery. The introduction of power perpetuals could also attract new participants to the cryptocurrency market, fostering a more vibrant and diverse financial ecosystem. As the adoption of power perpetuals grows, the emergence of network effects may further bolster the utility and value of the ecosystem, creating a virtuous cycle of growth and innovation.

Our journey begins with the launch of BTC2 and USDT3, but we have an ambitious roadmap to expand our offerings across a wide range of assets and markets. We encourage you to consult our product roadmap to stay informed about upcoming releases and exciting developments in the pipeline.

At Squared Labs, we highly value insights, ideas, and feedback from our community. We believe that open dialogue and collaboration are essential for shaping the future of power perpetuals and driving the growth of our ecosystem. We invite you to reach out to us on Discord to contribute to the discussion and share your thoughts. For more private conversations or specific inquiries, please feel free to open a support ticket on our Discord server. Your input will be instrumental in guiding the evolution of power perpetuals and the Squared Labs ecosystem as we work together to push the boundaries of what's possible in the world of decentralized finance.